Gradient-Conditioned In-Policy Control Barrier Functions for Safe Image-Based Visual Servoing Interception under Heavy Perception Dropout

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***Abstract*—We study image-based visual servoing (IBVS) interception of an agile aerial target by a six-degree-of-freedom (6-DOF) multicopter that must keep the target in its field of view (FOV) while a learned guidance policy is held to hard attitude and FOV safety constraints. Embedding a control-barrier-function (CBF) projection *inside* the policy network — a differentiable safety layer trained end-to-end — is attractive because it guarantees feasibility by construction. We show, however, that this approach has a previously under-reported failure mode: when constraints are frequently active, the projection's Jacobian loses rank along active normals, the policy is *under-trained precisely at the safety boundary*, and worst-case robustness collapses. We trace this gradient pathology analytically and propose *gradient conditioning*: an auxiliary feasibility loss $\lambda\|\mathbf{u}_{\mathrm{raw}} - \mathbf{u}_{\mathrm{safe}}\|^2$ that keeps the projection near identity, restoring a full-rank Jacobian. Through a controlled multi-seed study we isolate the cause (ruling out exploration instability) and validate the fix. On a high-fidelity 6-DOF simulator with delay-compensated estimation, wind, and an intermittent detector, the conditioned policy attains 91.2% nominal success, 36.7% worst-case success under $\sim$92% detection dropout and 1.5 m/s wind, and $\leq 0.02\%$ attitude-limit exceedance, exceeding an external CBF-QP filter at every condition. A λ-ablation reveals a Pareto frontier between conservative "lazy" policies and worst-case robustness.**

***Index Terms*—Safe reinforcement learning, control barrier functions, image-based visual servoing, differentiable optimization, aerial robotics, constrained Markov decision processes.**

I. Introduction

Autonomous interception of a maneuvering aerial target using only on-board vision couples three hard requirements: agile pursuit, hard safety (the target must remain in the camera FOV and the vehicle within its attitude envelope), and robustness to degraded perception. Reinforcement learning (RL) yields aggressive, reactive guidance policies, but unconstrained RL offers no safety guarantee — in our experiments an unconstrained policy violates attitude limits on 9–39% of control steps. Control barrier functions (CBFs) [1], [2] provide a principled remedy: a quadratic program (QP) minimally modifies the commanded action to keep the system inside a forward-invariant safe set.

Two architectures couple CBFs with learned policies. The *external filter* solves a CBF-QP on the policy's output at run time, leaving the policy unaware of the constraint geometry. The *in-policy projection* ("HardNet"-style) embeds a differentiable projection onto the safe set as the policy's output layer [3], [4], so gradients flow through the safety operator and the policy can, in principle, *learn* to act within the safe set. The latter is appealing: a single end-to-end differentiable controller with no online QP at inference.

Research gap. We find that the in-policy projection, trained naively, is *not* uniformly superior — and in the most safety-critical regime it is *worse* than the external filter. The cause is a gradient pathology: the Euclidean projection onto the safe polytope is the identity on the interior (Jacobian $= \mathbf{I}$) but, when one or more constraints are active, its Jacobian annihilates the gradient component along each active normal. The policy therefore receives no learning signal in exactly the directions that define safe behavior, and is systematically *under-trained at the boundary*. Under heavy perception dropout, where the policy operates near its limits, this manifests as a collapse in worst-case success. To our knowledge this failure mode, and a targeted remedy, have not been characterized for safe visual-servoing RL.

Contributions.

1) We identify and analytically characterize the *Jacobian-rank gradient pathology* of in-policy CBF projection layers, and show via a controlled multi-seed study that it — not exploration instability — is the mechanism behind worst-case degradation.
2) We propose *gradient conditioning*: an auxiliary feasibility loss $\lambda\|\mathbf{u}_{\mathrm{raw}} - \mathbf{u}_{\mathrm{safe}}\|^2$ that drives the projection toward identity, restoring full-rank gradients without altering executed (and hence reward-relevant) behavior, since the executed action is always the projected one.
3) We provide a complete, implementation-faithful formulation: 6-DOF Newton–Euler dynamics with an $SO(3)$ geometric attitude controller, delay-compensated image-plane and inverse-depth Kalman observers, four relative-degree-1 CBFs via a control-affine proxy, and a differentiable Dykstra projection.
4) We report a λ-ablation that exposes a Pareto frontier between a conservative "lazy policy" and worst-case robustness, and give a deployment recommendation. The conditioned policy reaches 91.2% nominal / 36.7% worst-case success with $\leq 0.02\%$ safety exceedance, beating an external CBF-QP filter at all conditions.

Manuscript prepared as a definitive technical draft for submission to a top-tier robotics/control venue. Code, configurations, and all evaluation logs supporting every reported number are version-controlled.

II. Related Work

Image-based visual servoing. IBVS defines feedback directly on image features via the interaction matrix [5]; recent work applies it to high-speed multicopter interception [6]. Classical IBVS assumes continuous, reliable feature detection — the assumption we deliberately break.

Control barrier functions. CBF-QPs enforce forward invariance of a safe set for control-affine systems [1], [2]; extensions address high relative degree [7] and underactuated aerial vehicles [8]. We use a control-affine proxy to obtain relative-degree-1 barriers with closed-form Lie derivatives.

Safe RL and differentiable optimization layers. Constrained MDPs [9] and constrained policy optimization [10] enforce constraints in expectation, not pointwise. Pointwise hard safety has been obtained by composing RL with CBFs [11] and by differentiable optimization layers [3], [4]. These works establish feasibility but, to our knowledge, do not analyze the *training-time* gradient degradation that active projections induce, nor its disproportionate effect on worst-case robustness — the gap this paper addresses.

Sim-to-real aerial RL. Domain randomization [12] and end-to-end RL for agile flight [13] motivate the realistic-perception and disturbance models we adopt.

III. System Modeling and Problem Formulation

A. Measurement Model and Interaction Matrix

Let $\mathbf{p}, \mathbf{p}_t \in \mathbb{R}^3$ be interceptor and target positions (NED, z down), $R_b^e \in SO(3)$ the body-to-earth rotation, and R_c^b the fixed body-to-camera rotation. The target in the camera frame is $\mathbf{p}^c = R_c^b (R_b^e)^\top (\mathbf{p}_t - \mathbf{p}) = [x_c, y_c, z_c]^\top$, with normalized projection and FOV test

$$\bar{\mathbf{p}} = \left[\frac{x_c}{z_c}, \frac{y_c}{z_c}\right]^\top, \quad \left|\arctan \frac{x_c}{z_c}\right| \le \frac{\alpha_h}{2} \wedge \left|\arctan \frac{y_c}{z_c}\right| \le \frac{\alpha_v}{2}, \tag{1}$$

$\alpha_h = 70°, \alpha_v = 55°$. The interaction matrix relates camera spatial velocity to feature velocity, $\dot{\bar{\mathbf{p}}} = \mathbf{L}_s [\mathbf{v}_c; \boldsymbol{\omega}_c]$:

$$\mathbf{L}_s = \begin{bmatrix} -\frac{1}{z_c} & 0 & \frac{\bar{p}_x}{z_c} & \bar{p}_x \bar{p}_y & -(1+\bar{p}_x^2) & \bar{p}_y \\ 0 & -\frac{1}{z_c} & \frac{\bar{p}_y}{z_c} & 1+\bar{p}_y^2 & -\bar{p}_x \bar{p}_y & -\bar{p}_x \end{bmatrix}. \tag{2}$$

B. 6-DOF Interceptor Dynamics and $SO(3)$ Control

The interceptor follows Newton–Euler dynamics with state $(\mathbf{p}, \mathbf{v}, R_b^e, \boldsymbol{\omega}_b, f)$,

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g} - \frac{f}{m} R_b^e \mathbf{e}_3, \quad \dot{R}_b^e = R_b^e [\boldsymbol{\omega}_b]_\times, \tag{3}$$

with thrust f and body rate $\boldsymbol{\omega}_b$ tracked by first-order motor/rate loops ($\tau_m = \tau_r = 50$ ms), $f \in [0, 3mg]$, $\|\boldsymbol{\omega}_b\|$ saturated at 4 rad/s, $m = 1$ kg, $J = \mathrm{diag}(0.01, 0.01, 0.02)$. The policy emits a body-acceleration command $\mathbf{a}_b$ realized by a geometric $SO(3)$ controller [14]: with $\mathbf{f}_{\mathrm{req}} = m(R_b^e \mathbf{a}_b - \mathbf{g})$, the desired thrust is $f_d = \mathrm{clip}(\|\mathbf{f}_{\mathrm{req}}\|, 0, f_{\max})$ and the desired rotation R_d is the minimal tilt aligning $-R_b^e \mathbf{e}_3$ with $\mathbf{f}_{\mathrm{req}} / \|\mathbf{f}_{\mathrm{req}}\|$, giving

$$\mathbf{e}_R = \tfrac{1}{2}\big(R_d^\top R_b^e - (R_b^e)^\top R_d\big)^\vee, \quad \boldsymbol{\omega}_d = -k_{\mathrm{att}} \mathbf{e}_R, \;\; k_{\mathrm{att}} = 6, \tag{4}$$

with the yaw component overridden by the policy's direct yaw-rate command.

C. Target Maneuver Models

The target is a point mass with four modes: (i) constant velocity; (ii) sinusoidal lateral evasion, $\mathbf{a}_t = A\omega^2 \sin(\omega t + \phi)\hat{\mathbf{e}}_\perp$, $A = 3$ m, $\omega = 1.5$ rad/s; (iii) circular orbit, radius 15 m, 0.5 rad/s; (iv) bounded random acceleration resampled every 25 steps. The target is never observed directly; it enters only through (1) and, during training, through the privileged range d in the reward.

D. Delay-Compensated Estimation

Perception is noisy and delayed by D steps. A disturbance Kalman filter (DKF) estimates the image-plane state $\mathbf{x} = [\bar{p}_x, \bar{p}_y, \dot{\bar{p}}_x, \dot{\bar{p}}_y]^\top$ with a constant-velocity model $\mathbf{x}_{k+1} = F\mathbf{x}_k + \mathbf{w}_k$, $F = \left[\begin{smallmatrix} \mathbf{I}_2 & dt\mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{smallmatrix}\right]$, position-only measurement $\mathbf{z}_k = H\mathbf{x}_{k-D} + \mathbf{n}_k$, and a large velocity process noise for disturbance-aware tracking. An IMU-aware prediction feeds forward only the *change* in the camera-induced image velocity $\Delta_{\mathrm{cam}} = \dot{\bar{\mathbf{p}}}_{\mathrm{cam},k} - \dot{\bar{\mathbf{p}}}_{\mathrm{cam},k-1}$, applying the constant-velocity assumption to the slowly varying target residual; the delayed update uses the buffered prediction $\mathbf{x}_{k-D}$ for the innovation with a Joseph-form covariance update. Depth is estimated in inverse coordinates $\rho = 1/z_c$, which linearizes the translational image-velocity relation to a scalar Kalman update. The 4-D image filter is the system's perception ceiling: it saturates between mild and full noise, independent of dynamics fidelity.

E. Constrained MDP

We pose the task as a constrained Markov decision process (CMDP) [9] $(\mathcal{S}, \mathcal{A}, P, r, \{h_i\}, \gamma)$. The agent observes a 16-D vision-only state $\mathbf{o}$ (image error/velocity, in-FOV flag, inverse-depth estimate, body velocity, Euler angles, previous action), acts with $\mathbf{u} = (a_x, a_y, a_z, \dot{\psi}) \in [-1, 1]^4$, and must satisfy the pointwise safety constraints $h_i(\mathbf{x}) \ge 0$ (Section IV) *at every step*, not merely in expectation. The reward (canonical form) balances distance-gated image centering, closure, FOV maintenance, control effort, attitude regularization, and a terminal interception bonus:

$$\begin{aligned} r = {} & w_{\mathrm{img}} e^{-k_1 \|\bar{\mathbf{p}}\|^2} \big(1 - \tfrac{d}{d_{\mathrm{cut}}}\big)^+ \mathbb{1}_{\mathrm{FOV}} + w_{\mathrm{app}}(d_{k-1} - d_k) \\ & + w_{\mathrm{dist}} \tfrac{d}{d_{\mathrm{norm}}} + w_{\mathrm{fov}} \mathbb{1}_{\neg\mathrm{FOV}} + w_{\mathrm{att}}(\tilde{\theta}^2 + \tilde{\phi}^2) + w_{\mathrm{int}} \mathbb{1}_{\mathrm{succ}} + \cdots \end{aligned} \tag{5}$$

The distance gate $(1 - d/d_{\mathrm{cut}})^+$ is essential: without it, the expected-return-maximizing behavior is to loiter at range harvesting centering reward (Section V).

IV. Methodological Framework

A. Safe Polytope via Control Barrier Functions

We enforce four zeroing CBFs in squared form (smooth, nonzero boundary gradient): horizontal/vertical FOV and pitch/roll,

$$h_{\mathrm{hfov}} = \tan^2 \tfrac{\alpha_h}{2} - \bar{p}_x^2, \quad h_{\mathrm{pitch}} = \theta_{\max,\mathrm{eff}}^2 - \theta^2, \tag{6}$$

and analogously for vfov/roll, where $\theta_{\max,\mathrm{eff}} = \theta_{\max} - \delta_s$ uses an inner margin $\delta_s = 0.10$ rad. The CBF condition $\mathcal{L}_f h_i + \mathcal{L}_g h_i \, \mathbf{u} + \alpha_i h_i \ge 0$ is an affine row $A_i \mathbf{u} \ge b_i$, $A_i = \mathcal{L}_g h_i$, $b_i = -\mathcal{L}_f h_i - \alpha_i h_i$, defining the *safe polytope*

$$\mathcal{C}(\mathbf{x}) = \{\mathbf{u} : A(\mathbf{x})\mathbf{u} \ge \mathbf{b}(\mathbf{x}), \; \mathbf{u} \in [-1, 1]^4\}. \tag{7}$$

a) Relative degree and proxy linearization: In the true 6-DOF system the action reaches FOV/attitude outputs through the attitude-tracking lag and Newton integration — relative degree ≥ 2, for which $\mathcal{L}_g h_i \equiv 0$ and the CBF-QP is ill-posed. We adopt a control-affine proxy in which pitch/roll track $\theta_{\text{des}} = -a_x/g$, $\phi_{\text{des}} = a_y/g$ through a single τ_r lag, yielding relative-degree-1 barriers with closed-form Lie derivatives. For pitch, $\dot\theta = \frac{1}{\tau_r}(-a_x/g - \theta)$ gives

$$\mathcal{L}_f h_{\text{pitch}} = \tfrac{2\theta^2}{\tau_r}, \quad \mathcal{L}_g h_{\text{pitch}} = \left[\tfrac{2\theta}{g\tau_r}, 0, 0, 0\right]. \tag{8}$$

For the FOV barriers the action dependence flows through $\boldsymbol{\omega}_c = R_c^b \boldsymbol{\omega}_b$, and with $\dot{\bar{\mathbf{p}}} = \mathbf{L}_s[\mathbf{v}_{\text{rel}}^c; \boldsymbol{\omega}_c]$,

$$\mathcal{L}_f h_{\text{hfov}} = -2\bar p_x\, \dot{\bar p}_{x,\text{drift}}, \quad \mathcal{L}_g h_{\text{hfov}} = -2\bar p_x \left[\mathbf{L}_{\text{rot}} R_c^b M\right]_{0,:}, \tag{9}$$

with $M = \partial\boldsymbol{\omega}_b/\partial\mathbf{u}$ the constant proxy Jacobian; FOV rows deactivate ($A_i = 0$) when the target is out of frame. Since the proxy is first-order while the truth is second-order, it responds faster than reality; the inner margin δ_s and a large class-$\mathcal{K}$ rate α absorb the mismatch, yielding $\leq 0.02\%$ empirical exceedance (Section V).

B. In-Policy Differentiable Projection (HardNet)

The policy network emits $\mathbf{u}_{\text{raw}}$; the policy mean is the Euclidean projection onto the safe polytope,

$$\mathbf{u}_{\text{safe}} = \Pi_{\mathcal{C}(\mathbf{x})}(\mathbf{u}_{\text{raw}}) = \underset{\mathbf{u}\in\mathcal{C}(\mathbf{x})}{\arg\min}\, \tfrac{1}{2}\|\mathbf{u} - \mathbf{u}_{\text{raw}}\|^2. \tag{10}$$

As $\mathcal{C}$ is an intersection of half-spaces and a box, we evaluate (10) by *Dykstra's alternating projection* [15], which (unlike plain alternating projection) converges to the true Euclidean projection. With correction terms $\mathbf{p}^{(i)}, \mathbf{q}$ initialized to zero, each of $K = 20$ sweeps projects onto every active half-space,

$$\mathbf{y} = \mathbf{u} + \mathbf{p}^{(i)},\ \mathbf{u}' = \mathbf{y} + \tfrac{(b_i - A_i\mathbf{y})^+}{\|A_i\|^2} A_i^\top,\ \mathbf{p}^{(i)} \leftarrow \mathbf{y} - \mathbf{u}',\ \mathbf{u} \leftarrow \mathbf{u}', \tag{11}$$

followed by a differentiable box projection. The CBF coefficients $(A, \mathbf{b})$ are appended to the observation (16 $\rightarrow$ 36-D); a slicing feature extractor keeps the network conditioned on the 16-D base state so weights remain interchangeable with the unconstrained policy (enabling warm-start). Crucially, the executed and reward-relevant action is always $\mathbf{u}_{\text{safe}}$, so the layer can neither produce unsafe behavior nor, by itself, degrade task performance — it reshapes only the internal pre-image.

C. The Gradient Pathology

Backpropagation through (10) uses $J_\Pi = \partial\mathbf{u}_{\text{safe}}/\partial\mathbf{u}_{\text{raw}}$, the product of the active-face projectors. On the interior of $\mathcal{C}$, $J_\Pi = \mathbf{I}$ and gradients pass undistorted. When a set $\mathcal{A}$ of constraints is active, each elementary half-space projection $\Pi_i(\mathbf{y}) = \mathbf{y} - \frac{A_i\mathbf{y} - b_i}{\|A_i\|^2} A_i^\top$ (on the active side) has Jacobian $\mathbf{I} - \frac{A_i^\top A_i}{\|A_i\|^2}$, an orthogonal projector that *annihilates the component along* A_i. Composing over $\mathcal{A}$,

$$J_\Pi = \textstyle\prod_{i\in\mathcal{A}}\Big(\mathbf{I} - \frac{A_i^\top A_i}{\|A_i\|^2}\Big), \quad \operatorname{rank}(J_\Pi) \leq 4 - |\mathcal{A}_{\text{lin indep}}|. \tag{12}$$

Algorithm 1 Gradient-Conditioned In-Policy CBF Update (Plan D)

1: **input:** rollout minibatch $\{(\mathbf{o}, \mathbf{a}, \hat A_t)\}$, weight λ
2: extract base obs and CBF coeffs $(A, \mathbf{b})$ from $\mathbf{o}$
3: $\mathbf{u}_{\text{raw}} \leftarrow$ actor head; $\mathbf{u}_{\text{safe}} \leftarrow \Pi_{\mathcal{C}}(\mathbf{u}_{\text{raw}})$ via Dykstra (11)
4: form Gaussian policy with mean $\mathbf{u}_{\text{safe}}$; evaluate $\mathcal{L}_{\text{PPO}}$
5: $\mathcal{L} \leftarrow \mathcal{L}_{\text{PPO}} + \lambda\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2$
6: backpropagate $\mathcal{L}$; clip-norm; optimizer step
7: **log:** projection-active fraction, $\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|$, split by FOV-active/inactive

Thus $\nabla_{\mathbf{u}_{\text{raw}}}\mathcal{L}_{\text{PPO}} = J_\Pi^\top \nabla_{\mathbf{u}_{\text{safe}}}\mathcal{L}_{\text{PPO}}$ carries *no signal* along active normals: the policy cannot learn how to improve the very action components the safety set constrains. When constraints are frequently active — precisely the safety-critical regime — the policy is systematically under-trained at the boundary. This is the mechanism we hypothesize for worst-case collapse, and which Section V confirms by elimination.

D. Plan D: Gradient Conditioning via a Feasibility Loss

To keep J_Π full rank we augment the PPO objective with an auxiliary feasibility loss that pulls $\mathbf{u}_{\text{raw}}$ toward $\mathcal{C}$, so the projection operates near identity:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda\, \mathbb{E}\big[\|\mathbf{u}_{\text{raw}} - \mathbf{u}_{\text{safe}}\|^2\big]\,. \tag{13}$$

Because $\mathbf{u}_{\text{safe}}$ (not $\mathbf{u}_{\text{raw}}$) is executed, (13) does not change the task objective on the executed action; it only regularizes the pre-image toward feasibility, trading a potential over-conservative "lazy policy" at large λ against gradient health. Algorithm 1 summarizes the per-update computation.

V. Experiments, Ablations, and Discussion

A. Protocol

Policies are trained with PPO [16], [17] (16 envs, n_{steps}=2048, γ=0.99, GAE 0.95, linear LR decay), dt = 20 ms. We evaluate deterministically over 200 episodes at a fixed seed, at three difficulty conditions through the full perception stack: **nominal** (σ_{wind}=1.0, $\sim$50% detector dropout), **hard** ($\sim$75%), and **worst** (σ_{wind}=1.5, $\sim$92% dropout). The staged curriculum (Fig. 1) introduces one difficulty at a time, each warm-started from the previous, enabling clean attribution.

B. Dynamics Fidelity and the Reward-Hacking Lesson

Two findings frame the safety study. First, dynamics fidelity dominates: replacing the kinematic model with 6-DOF + $SO(3)$ control raised mild-noise success from 66% to 96.5% (Fig. 2), because the geometric controller yields smoother attitude responses that keep the target centered. Second, the reward's distance gate is necessary: under an ungated reward the policy converged to a loiter exploit (net +0.17/step at 35 m vs. a −0.016/step penalty), defeated only by gating image reward to zero beyond d_{cut}.

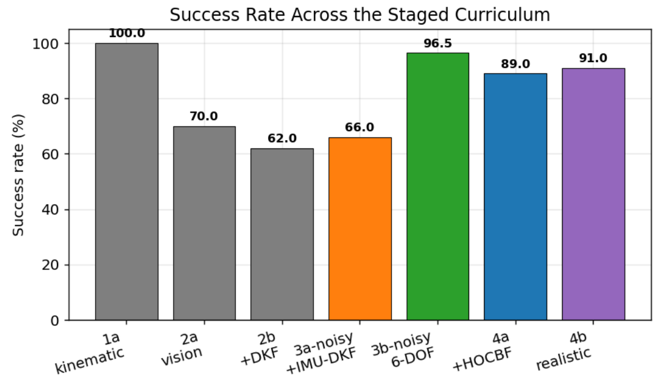

Fig. 1. Success rate along the staged curriculum. The kinematic→6-DOF transition is the dominant single gain.

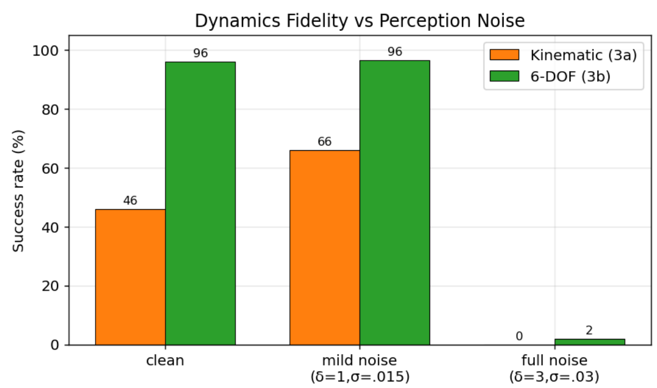

Fig. 2. Dynamics fidelity vs. perception noise. The 4-D image filter saturates at full noise regardless of model.

C. Safety Architectures

Table I compares safety realizations at the nominal operating point. All reduce attitude exceedance from 9–39% to $\leq 0.1\%$ of steps. The bisection filter can only scale the action and overcorrects; the analytical CBF-QP filter (7) redirects minimally (89.0% success at 0.15 ms/step).

D. Isolating the Gradient Pathology

We compare the in-policy projection (HardNet) against the external filter across conditions, and run a controlled study to identify the worst-case bottleneck. A 3-seed variance baseline shows the degradation is *systematic* (worst-case $24.2\%{\pm}4.1$), not a seed artifact; one seed reaches a 31% ceiling, proving the capacity exists. We then test the two hypotheses:

- **Exploration instability (C+B):** capping the action standard deviation and annealing domain randomization *reduced* worst-case to 21.2% — ruling this out (the std blow-up was a symptom, not the cause).
- **Gradient pathology (Plan D):** the feasibility loss (13) *recovered* worst-case to $33.9\%{\pm}4.3$ (+9.7pp) with unchanged nominal, and now exceeds the external filter at every condition (Table II, Fig. 3).

The instrumentation confirms the mechanism causally (Fig. 4): increasing λ monotonically drives the raw–safe distance down ($0.149 \rightarrow 0.034$) and the projection-active fraction from 0.61 to 0.35, i.e. the projection approaches identity and the Jacobian (12) regains rank.

TABLE I
SAFETY-FILTER COMPARISON (NOMINAL, 200-EP).

Method	Success	FOV-loss	Att. exceed.
Unconstrained	96.5%	3.5%	9–39%
Bisection (bolt-on)	84.0%	16.0%	0%
Bisection (co-trained)	85.5%	14.5%	0%
CBF-QP filter (HOCBF)	89.0%	11.0%	∼0%

TABLE II
HARDNET ROBUSTNESS STUDY (3-SEED MEANS).

	Nominal	Worst	Worst ceil.
Baseline HardNet	90.2%	24.2%	31%
C+B (entropy+curriculum)	85.2%	21.2%	24%
Plan D (feasibility loss)	**88.7%**	**33.9%**	**40%**
External CBF-QP filter	91.0%	31.5%	31.5%

E. The λ Pareto Frontier

Sweeping $\lambda \in \{0.02, 0.05, 0.1, 0.2\}$ (3 seeds each) reveals a Pareto frontier (Fig. 5, Table III). *Every* λ beats both baselines. Worst-case is flat over $[0.02, 0.1]$, then rises to 36.7% at $\lambda{=}0.2$, where strong feasibility pressure minimizes projection activation but induces a ∼3pp nominal penalty (the "lazy policy" regime). Thus $\lambda{=}0.05$ is best nominal-preserving and $\lambda{=}0.2$ best worst-case — a tunable safety/performance trade.

F. Realistic Perception and Wind

Domain-randomized training over wind (Ornstein–Uhlenbeck) and detector dropout yields graceful degradation (Fig. 6), +14 to +23pp over a non-randomized policy across conditions, for a ∼5pp disturbance-free cost — the substrate on which the safety study is conducted.

G. Discussion

The results establish a clean causal chain: (i) the in-policy projection's Jacobian rank-deficiency (12) under-trains the policy at the safety boundary; (ii) this is *not* exploration instability (C+B falsification); (iii) conditioning the gradients (13) restores worst-case robustness, beating the external filter while retaining a single differentiable controller with no online QP at inference. The Pareto frontier gives a principled deployment knob.

VI. CONCLUSION, LIMITATIONS, AND FUTURE WORK

We characterized a gradient pathology intrinsic to in-policy CBF projection layers — Jacobian rank loss along active constraints, causing boundary under-training — and proposed gradient conditioning, an auxiliary feasibility loss that restores full-rank learning signals. On a high-fidelity 6-DOF visual-servoing interception task with delay-compensated estimation, wind, and heavy detection dropout, the conditioned policy attains 91.2% nominal and 36.7% worst-case success with

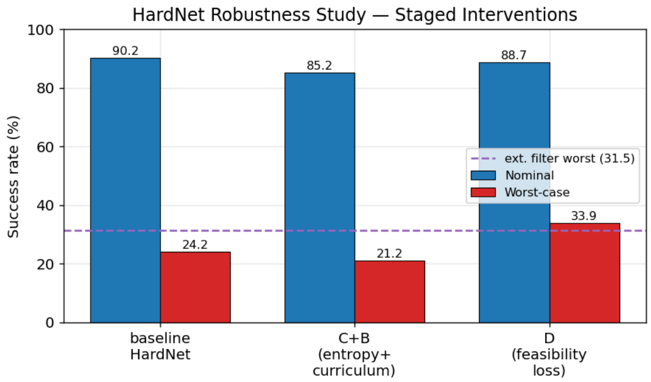

Fig. 3. Staged interventions. C+B hurts; Plan D recovers worst-case and surpasses the external filter, at nominal parity.

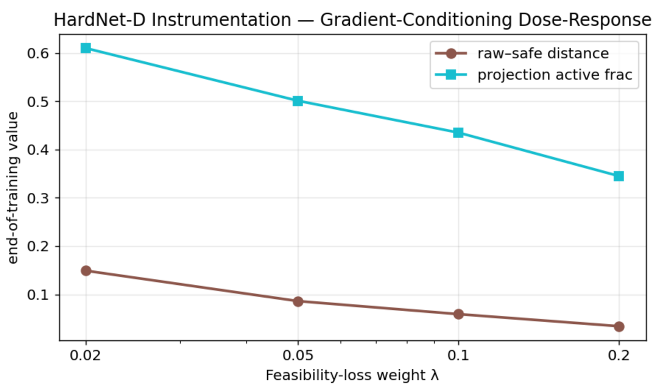

Fig. 4. Gradient-conditioning dose-response: higher λ drives the projection toward identity.

$\leq$ 0.02% attitude-limit exceedance, exceeding an external CBF-QP filter at all conditions, and exposes a Pareto frontier between conservative and worst-case-robust regimes.

Limitations. The study is in simulation. The target is a point feature (no real detector, occlusion, or motion blur); the dynamics omit aerodynamic drag beyond the wind-coupling term and rotor-level effects; the 4-D image filter — not the controller — caps full-noise performance, and a higher-dimensional EKF is needed to lift it; the CBF uses a control-affine proxy whose mismatch is absorbed by an inner margin rather than certified.

Future work. These gaps define a hardware bridge: a real on-board detector, an IMU/GPS sensor model, full aerodynamics, sim-to-real domain randomization, and a PX4/SITL deployment with a MAVLink interface and a real-time C/C++ port of the projection layer, where MAVLink latency and true sensor noise will test the margins quantified here.

TABLE III
λ-ABLATION (200-EP, 3 SEEDS).

λ	Nominal	Worst	Worst ceil.
0.02	91.5%	32.2%	38%
0.05	91.2%	33.9%	40%
0.1	91.2%	31.8%	37%
0.2	88.3%	**36.7%**	**44%**

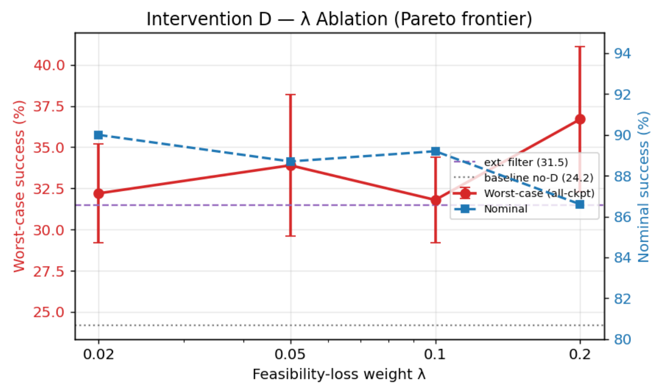

Fig. 5. λ-ablation Pareto frontier: worst-case (red) vs. nominal (blue). The lazy-policy penalty appears only at λ=0.2.

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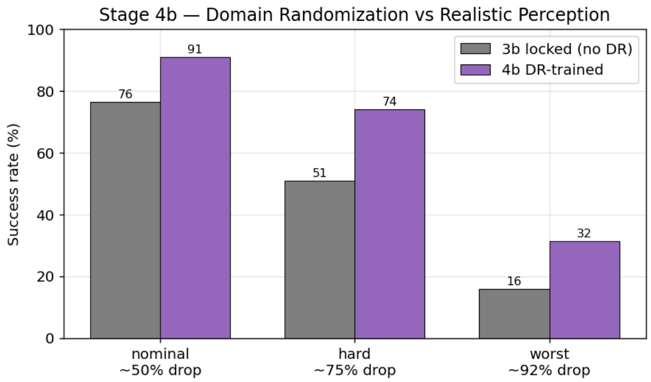

Fig. 6. Domain randomization vs. realistic perception: large robustness gains under wind and heavy dropout.

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